

Quantum SVM Error Report

Error Message:

`qiskit.transpiler.exceptions.TranspilerError: 'Number of qubits (397) in circuit-158952 is greater than maximum (28) in the coupling_map'`

Cause:

This error occurs because the input data, after feature extraction and encoding, results in quantum circuits with 397 qubits. The target quantum backend (or simulator) only supports a maximum of 28 qubits. This often happens when using TF-IDF or other vectorization techniques that generate a high number of features.

Problems Identified: TF-IDF generated too many features (dimensions). Quantum Kernel method requires a circuit with the same number of qubits as input features. The quantum backend (like AerSimulator or real hardware) has a strict limit on qubits. **Possible Fixes: Dimensionality**

Reduction: Apply feature reduction methods like PCA, TruncatedSVD (for sparse data), or manually reduce TF-IDF dimensions. **Limit Features:** Limit `max_features` in `TfidfVectorizer`, e.g., `TfidfVectorizer(max_features=20)`. **Custom Encoding:** Use a more compact encoding method for text input suitable for quantum circuits, like binary or amplitude encoding. **Use fewer**

symptoms or domain knowledge: If applicable, filter out symptoms/features with low importance. **Recommendation:**

Start with limiting TF-IDF to 10–20 features and test if the quantum kernel works. Gradually increase until hitting backend limits.